AREC 261 · Module 1 Test — SAMPLE

Dataset for this test: Saskatchewan RM Crop Yields. You are provided two files: the wide file `rm_yields_1990plus.csv` (one column per crop) and the long file `rm_yields_1990_2025.csv` (crop name in its own `Crop` column, for PivotTables). Lentils are recorded in pounds per acre; every other crop is in bushels.

Submission. Submit one `.xlsx` workbook with a separate, clearly labelled worksheet for each question. For each question, 20% of the mark is for presentation (labelled cells, sensible formats, a reader can follow the sheet) and 80% for correctness.

Question 1

Copy this table into a blank sheet: select it, copy, and paste into cell A1.

Field	Acres	Yield (bu/ac)
Kestrel	120	41.2
Meadowvale	240	38.6
Nightjar	95	44.8
Home	310	36.1
Rented	175	39.9

Off to the right of the table, in a column with nothing else in it, put the canola price **$14.20** in a single labelled cell.

(a) Add a column that calculates revenue per acre (yield X price) for each field using relative and absolute references. Format as currency with two decimals.

(b) Add another column that calculates total revenue per field. Format as currency a thousands separator and no decimals.

(c) Under the table, use `SUM` to total the acres, the total bushels (yield × acres), and the total revenue.

(d) Explain why we need to include a `$` in the price reference in part a but not in part b.

Question 2

This question uses `rm_yields_1990plus.csv`.

Filter the data to 2023. Then copy the **Canola** yields into a new worksheet.

(a) Compute the mean and the median canola yield.

(b) Compare the two values. What does the relationship between the mean and the median tell you about how yields were distributed across RMs that year?

(c) Compute the standard deviation and the interquartile range (IQR) of the same column.

(d) Would the mean or the median better represent a typical RM here? Justify your choice using your answers to parts a-c.

Question 3

This question uses `rm_yields_1990plus.csv`.

(a) Use `COUNTIFS` to count how many RM-year rows have **canola above 40 AND spring wheat above 45** in the same row.

(b) Use `COUNTIF` to count rows with canola above 40 (ignoring wheat).

(c) Explain why the count in (a) is smaller than the count in (b).

Question 4

This question uses `rm_yields_1990_2025.csv` (the long file).

(a) Build a PivotTable showing each RM's spring wheat yield in 2023, sorted from highest to lowest.

(b) In a cell beside the table, report the three highest-yielding RMs and their yields.